

L13b: Interpretable Allocation and Model-Predictive Rebalancing

CHEME 5660 Cornell University

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Objectives for Today

Previously, we chose the share counts of a portfolio once. How do we manage that portfolio every day, and how do we judge it when the real market gives us only one path?

Today's concepts

- Allocation: maximum utility within a budget and trading limits, with a score for each stock that combines its expected growth rate with the market state.
- Rebalancing: a model-predictive controller that re-plans at the horizon, at a band exit, or at a drawdown stop.
- Evaluation: one historical path, and then typical and stressed synthetic markets.

What You Need to Know

Core concepts

- Each rebalance maximizes a utility of the stock scores within a budget and four trading limits. The score has the sign of the stock's expected growth rate, and the market state changes only its size.
- The method re-plans after 21 trading days, when the wealth leaves the 95% forecast band, or when it falls 8% below its peak.
- A backtest is one historical path. A market generator fitted to the same stocks gives many typical and stressed markets for testing the same method.

For understanding (you won't reproduce these)

- The four utilities, their solutions under the budget alone, and the three benchmarks.
- The forecast model, the market hedge, and the trading costs.
- The ten selection rules, and how the typical and stress sets are chosen.

Lectures and Examples

Lecture:

CHEME-5660-L13b-Lecture-MPC-Allocation-Fall-2026.ipynb

Examples:

CHEME-5660-L13b-Example-MPC-RealData-Fall-2026.ipynb

CHEME-5660-L13b-Example-MPC-SyntheticData-Fall-2026.ipynb

Live Portfolio Management

Live portfolio management couples three problems: allocation (what to hold), rebalancing (when to trade), and evaluation beyond the historical record. We'll treat the three problems as one system. We call the portfolio that this system manages the book.

The study uses daily data for 373 S&P 500 stocks. It calibrates on 2014 to 2024 and trades the 420 trading days from January 2025 to September 2026. Every decision uses only data available on the day of the decision.

Study: J. D. Varner, INFORMS Annual Meeting 2026, arXiv:2611.03022.

Allocation: The Score γ

A selection rule picks a basket $\mathcal{N}$ of K stocks, or names. A weighting rule then chooses the shares $n_i \geq 0$ of each stock i. The method gives each stock a score:

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max\left(|\beta_i|,\,0.1\right)^{\xi}}\right)$$

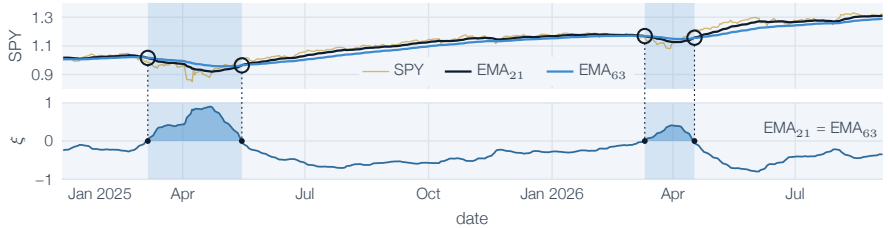
where α_i and β_i are the single index model parameters of stock i against `SPY`. The market growth rate $\tilde{g}_M$ is the 252-day EMA of the daily growth rate of `SPY` (units: 1/yr). The numerator is the expected growth rate, and the denominator is the β penalty.

The Market State

The penalty depends on the market state ξ, which compares the 21-day and the 63-day EMA of the SPY close:

$$\xi = -G\left(\frac{\text{EMA}_{21}}{\text{EMA}_{63}} - 1\right)$$

where the gain $G = 21.23$ keeps $|\xi| \leq 1$ on 99% of the days from 2014 to 2024:



The top strip shows SPY and its two EMAs, each divided by the first SPY close. The market state was positive on the shaded days, 75 of the 420 trading days.

Reading the Score

How do we read the score?

The score has the sign of the expected growth rate, because the β penalty is always positive.

In a falling market ($\xi > 0$), the penalty shrinks the scores of stocks with $|\beta_i| > 1$ toward zero and favors stocks with a low β. In a rising market ($\xi < 0$), the score favors stocks with a high β.

The hyperbolic tangent keeps the score between -1 and 1. The absolute value and the floor of 0.1 extend the preference weight we used previously to stocks with $\beta_i \leq 0$.

Maximize Utility Subject to Trading Limits

With close price S_i, wealth W, and weight $w_i = n_i S_i / W$, each rebalance chooses the shares $n = (n_1, \dots, n_K)$ to solve:

$\underset{n \geq 0}{\text{maximize}}$	$U(n; \gamma)$	(utility)
subject to	$\sum_{i \in \mathcal{N}} n_i S_i \leq W$	(budget)
	$w_i \leq \bar{w}$	(single-name cap)
	$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$	(volatility cap)
	$\bar{c} \, \lVert n - n^{\text{prev}} \rVert_1 \leq B$	(turnover budget)
	$\beta^\top w \leq \bar{\beta}$	(β cap, hedge on)

where n^{prev} holds the shares before the rebalance. Wealth that the budget leaves unspent stays in cash.

The Trading Limits

Limit	What it does	Setting in the study
Single-name cap $\bar{w}$	Each stock is at most the fraction $\bar{w}$ of the wealth.	$\max(0.10,\ 1.5/K)$, which is 11.5% for 13 stocks
Volatility cap σ_{SPY}	The book is no more volatile than `SPY`, where Σ_g is the covariance of the daily growth rates.	17.3% a year
Turnover budget B	A rebalance spends at most B on the half spread $\bar{c}$ per share traded.	$0.10\,W$, with a half spread of 5 basis points of the mean close price
β cap $\bar{\beta}$	With the market hedge on, the book's β to `SPY` is at most $\bar{\beta}$.	$\max(0,\ 1-\xi)$

With the hedge on, the book can also hold up to five stocks with $\beta_i \leq 0$.

The Weighting Rules

The study compares four utilities of the stocks with $\gamma_i > 0$:

Utility	$U(n;\gamma)$	Solution with the budget alone
Cobb–Douglas	$\sum_i \gamma_i \ln n_i$	$w_i \propto \gamma_i$
CES on shares	$\left(\sum_i \gamma_i \, n_i^{\rho}\right)^{1/\rho}$	$n_i \propto (\gamma_i / S_i)^{\eta}$
CES on weights	$\left(\sum_i \gamma_i \, w_i^{\rho}\right)^{1/\rho}$	$w_i \propto \gamma_i^{\eta}$
Stone–Geary	$\sum_i \gamma_i \ln (n_i - b_i)$	$(n_i - b_i)\, S_i \propto \gamma_i$

where $\rho = 1 - 1/\eta$ for the elasticity $\eta = 0.5$, and the baseline b_i is the number of shares worth 1% of the wealth. With the trading limits, a numerical solver finds the maximum.

Three benchmarks run under the same limits: minimum variance, the tangency portfolio, and a 130/30 long–short book that shorts stocks with $\gamma_i < 0$. The four utilities and the three benchmarks are the seven weighting rules.

Rebalancing: The Forecast Band

When should the book trade? Trading every day costs money. A fixed calendar ignores what the market does between trades.

The method uses model-predictive control. The controller forecasts the wealth of the book and measures the real wealth after each close. When the measurement departs from the forecast, the controller makes a new plan, called a re-plan.

At each re-plan, the controller simulates 1,000 price paths over the next 21 trading days with a multi-asset geometric Brownian motion. With $\mu_W(\tau)$ and $\sigma_W(\tau)$ the mean and standard deviation of the forecast wealth τ trading days after the re-plan, the 95% forecast band is:

$$\mu_W(\tau) - 1.96\,\sigma_W(\tau) \;\le\; W \;\le\; \mu_W(\tau) + 1.96\,\sigma_W(\tau)$$

The Rebalancing Triggers

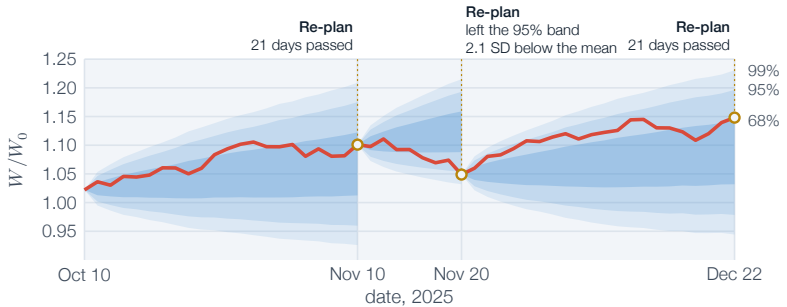
After each close, the controller checks three triggers in order and re-plans on the first trigger that holds:

1. Drawdown stop: the wealth is more than 8% below its running peak. The re-plan resets the peak to the current wealth.
2. Horizon: the plan has run for 21 trading days, where the forecast ends. The next forecast starts from the measured wealth.
3. Band exit: the wealth has left the 95% band. The book has gained or lost more than the model expected.

Each re-plan solves the same allocation problem from the shares held, with that day's estimates. Between re-plans, the book does not trade.

One Episode of the Real-Data Backtest

The red line is the wealth of the book divided by its starting wealth W_0. This book holds 13 hand-picked stocks, with the Cobb–Douglas utility and the hedge off:



The second re-plan came when the wealth fell 2.1 standard deviations below the forecast mean. From January 2025 to September 2026, this book re-planned 27 times: 17 at the horizon, 7 at a band exit, and 3 at a drawdown stop.

Fixed Schedule and Trading Costs

Do the triggers beat a fixed schedule?

On a fixed 21-day schedule, the ending wealth was the same. The triggers took about a point off the worst drawdown, on real prices and in simulated sell-offs.

What does a trade cost?

Each trade fills at the next open and pays a half spread of 5 basis points plus a market-impact cost. There is no commission.

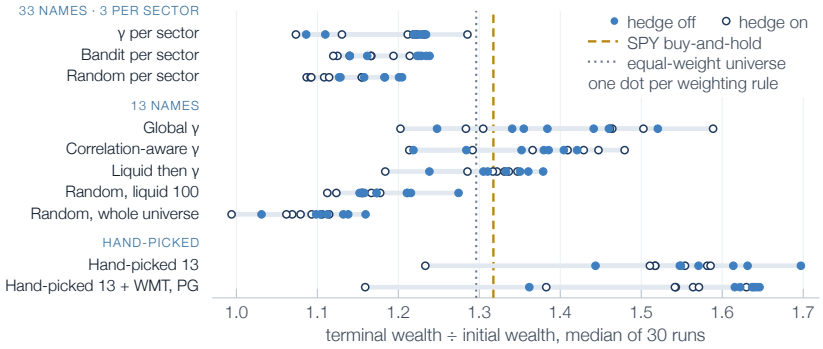
The Selection Rules

We ran every pair of ten selection rules and seven weighting rules on the real data, with the hedge on and off. The selection rules come in three groups:

- 33 stocks, 3 per sector: in each of the 11 sectors, the rule takes the 3 highest scores, the top 3 stocks of a bandit, or 3 stocks at random. The bandit learns which stocks in a sector grew faster than `SPY` over the trailing year.
- 13 stocks: three rules take the 13 highest scores, from the whole universe, from the 100 most traded stocks, or skipping a stock correlated above 0.7 with one already picked. Two random rules draw 13 stocks from the same two pools.
- Hand-picked: the book holds 13 stocks chosen by hand without 2025 data and never re-picks them. A second list adds WMT and PG.

Selection Matters More Than Weighting

Each row is a selection rule, and each dot is a weighting rule:



With the hedge off, seven weighting rules on the 33 stocks picked by score ended between 1.09 and 1.23 times the initial wealth. Different baskets ended between 1.03 and 1.70.

Example: The Method on Real Data

Example:
CHEME-5660-L13b-Example-MPC-RealData-Fall-2026.ipynb

In this example, we run one book day by day. We then change the weighting rule, the selection rule, the hedge, and the basket on the same path.

One path cannot tell us whether a rule was good or lucky.

Market Generator: JumpHMM

How do we judge the method beyond the historical record? Real data gives one path of prices, with one broad sell-off in the spring of 2025. A market generator gives many synthetic markets for the same stocks. The growth rate of synthetic stock i in sector s on trading day t is:

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector}} + \underbrace{\psi_i\, \tilde{\varepsilon}_i(t)}_{\text{own}}$$

where the market, sector, and own draws each come from a jump hidden Markov model (JumpHMM). The loadings δ_i and ψ_i scale the sector and own draws, and each $\beta_i(t)$ drifts around the stock's estimate. The generator is fitted to the same 373 stocks on 2014 to 2024. It reproduces the fat tails of real growth rates, but its volatility clustering fades after a day. The controller still forecasts with a geometric Brownian motion, which has no fat tails.

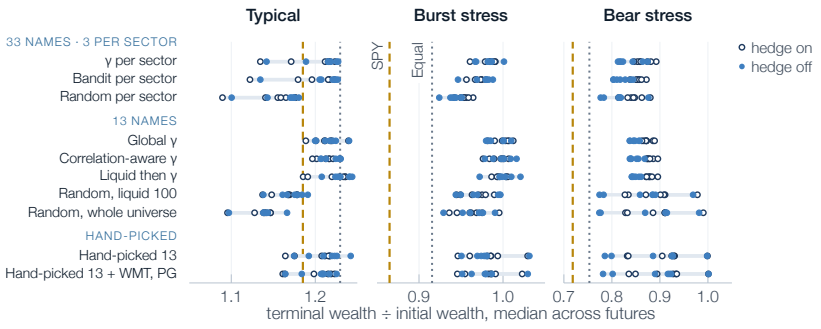
Typical and Stress Sets

On each synthetic market, the method calibrates on 2014 to 2024 and trades January 2025 to September 2026 with the real settings. We draw 10,000 markets and build three sets from them:

- Typical: this set holds 600 markets drawn at random. The median `SPY` drawdown is 17.6%.
- Burst stress: this set holds the 200 deepest `SPY` drawdowns among markets with a jump burst, months of large moves of both signs. The median `SPY` drawdown is 36.4%.
- Bear stress: this set holds the 200 deepest `SPY` drawdowns among markets without a burst, where prices fall more slowly. The median `SPY` drawdown is 40.6%.

Sell-Offs: Every Median Ends Above SPY

The rows, dots, and lines are as in the real-data figure. Each dot is the median across the markets of a set:



In both stress sets, every median ended above `SPY` and above the equal-weight portfolio. The random selection rules did as well.

Reading the Synthetic Results

- The margin: the margin over `SPY` came from the allocation, not from the timing of the re-plans.
- Selection by score: in typical markets with the hedge off, 13 stocks picked by score beat 13 random stocks in 62 to 71% of the markets. In the bear stress set, random picks did as well or better for most weighting rules. The generator keeps each stock's past trend, which favors selection by score more than real prices did.
- The hedge: the hedge reduced terminal wealth on real prices, but it added to terminal wealth in the bear stress set.

Example: The Method on Synthetic Markets

Example:
CHEME-5660-L13b-Example-MPC-SyntheticData-Fall-2026.ipynb

In this example, we run the method on markets from each set and compare the books with `SPY` and an equal-weight portfolio.

The example uses a few markets in each set, and its counts are rougher than the study's.

Summary

We followed one system for managing a stock portfolio every day, from the allocation problem to its test on real and synthetic markets.

Key takeaways

- In allocation, stock selection mattered more than the weighting rule. Seven weighting rules on one basket ended close together, while different baskets ended far apart.
- In rebalancing, re-plans came at the horizon, a band exit, or a drawdown stop. The triggers gave the same ending wealth as a fixed schedule and took about a point off the worst drawdown.
- In both sell-off sets, every median ended above `SPY`. The generator keeps each stock's past trend, which favors selection by score.

A paper-trading version of this system is running now.